Technical Notes: Verilog RTL and RISC-V Practice Activities

Day 3: Single-cycle RISC-V CPU

3.1 Instruction extensions

- **Approach:** Added ANDI, ORI, XORI, SLTI, SLLI, SRLI, LUI and JAL. The immediate generator now decodes I, S, B, U and J layouts, and the control unit has new decode entries for each opcode.
- **Verification:** Each instruction is exercised in coverage.hex and checked in cpu_tb.
- **Result:** All new instructions produce the expected register values (see the table in report/).
- **Issues:** No significant issues recorded. Design note: shift-immediate instructions take the shift amount from the rs2 field, not from a sign-extended immediate.

3.2 Assembly programs

- **Approach:** Hand-assembled (a) sum of 1 to 10, (b) the first 10 Fibonacci numbers stored to data memory, (c) maximum of an 8-word array. Each .hex line has a comment with the assembly.
- **Verification:** Expected values: x2 = 55; memory[0..9] = 0,1,1,2,3,5,8,13,21,34; x12 = 99.
- **Result:** All three programs produce the expected results.
- **Issues:** No significant issues recorded.

3.3 Self-checking CPU testbench

- **Approach:** cpu_tb.v loads each program, runs until the ECALL stop marker (with a cycle limit to avoid hangs), then compares registers or memory with expected values.
- **Verification:** One PASS or FAIL line per program.
- **Result:** All programs PASS.
- **Issues:** No significant issues recorded.

3.4 Waveform analysis

- **Approach:** Captured pc, instruction, alu_result and writeback_data for program (a).
- **Verification:** Annotated three points: an addi at PC 0x00, the accumulating add at 0x0c, and the bge at 0x14.
- **Result:** x2 accumulates 1 to 55. The branch is taken while x3 <= 10 and not taken when x3 reaches 11.
- **Issues:** None in the design. Status: the annotated screenshot still has to be captured from ModelSim and saved in day3/waves/ (cpu.vcd is the waveform data, not a screenshot).

3.5 Branch and jump extensions

- **Approach:** Added BNE, BLT, BGE (signed comparison) and JALR, which writes PC+4 to rd and jumps to (rs1 + imm) & ~1.
- **Verification:** Directed tests with signed positive, negative and zero operands, and a link-address check for JALR.
- **Result:** Passed.
- **Issues:** No significant issues recorded.

3.6 More assembly programs

- **Approach:** Bubble sort of 8 words, GCD by repeated subtraction, and multiplication by shift-and-add.
- **Verification:** Sorted memory contents and the GCD and product values compared in the testbench.
- **Result:** All three PASS.
- **Issues:** No significant issues recorded.

3.7 Instruction coverage matrix

- **Approach:** Table of every supported instruction against the test that exercises it.
- **Verification:** Added coverage.hex for instructions not hit by programs 1 to 3.
- **Result:** Every instruction in the matrix has at least one passing test (day3_coverage.md).
- **Issues:** No significant issues recorded.

3.8 Bug-hunt

- **Approach:** Three copies of the CPU, each with one injected bug: zero-extended negative I-immediate, writable x0, and branch target adding 4 twice.
- **Verification:** bug_hunt_tb runs a focused program against each copy.
- **Result:** All three bugs were detected: x1 = 0x00000ff8 instead of 0xfffffff8, x1 = 6 instead of 1, and the CPU still running at PC 0xA8 at the cycle limit. No new tests were needed.
- **Issues:** No significant issues recorded.